

FMD

FindMyDentist

Technical & Product Blueprint

A unified dental technology ecosystem spanning patients, dentists, clinics, education, research and AI.

Purpose: This document is the product/technical direction to share with the FMD development team. It describes what FMD should become, how the major products fit together, and the required AI architecture.

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1. FMD Vision

FMD should not be positioned as simply another dentist directory or booking marketplace. The long-term product is a **dentistry-first digital ecosystem** combining discovery, content, AI, patient continuity, clinic workflows, education and research.

Core product principle: One FMD identity and intelligence layer, with multiple specialized experiences.

Product	Primary users	Core purpose
FMD Main	Patients, dentists, students, public	Dental discovery, AI guidance, dentist profiles, content, booking, records and continuity
FMD Clinic	Dental clinics / dentists	Clinic management, appointments, reminders, records, billing and workflow
FMD Academic	BDS, MDS, faculty, researchers	AI seminars, PPTs, research, journal club, case presentations, posters and viva
Future modules	Institutions / ecosystem partners	FMD Campus and additional ecosystem products, introduced later

Current development priority: FMD Main + FMD Clinic. FMD Academic should be architected as a first-class product in the same ecosystem, with launch timing determined by development capacity.

2. Ecosystem Architecture

The user experience should feel like a family of products under one powerful brand, rather than unrelated apps. FMD Main, FMD Clinic and FMD Academic should share authentication, AI infrastructure, user identity, notifications, payments where appropriate, analytics and selected dental knowledge services.

Conceptual structure

FMD

- ■ ■ ■ FMD Main – public dental ecosystem
 - ■ ■ ■ AI Dental Assistant
 - ■ ■ ■ Dentist profiles / social content
 - ■ ■ ■ Explore / reels / posts
 - ■ ■ ■ Appointment discovery & booking
 - ■ ■ ■ Patient records / X-rays / photos
 - ■ ■ ■ Digital warranty / treatment continuity
- ■ ■ ■ FMD Clinic – dental practice operating layer
 - ■ ■ ■ Appointments & reminders
 - ■ ■ ■ Patient management
 - ■ ■ ■ Clinical uploads
 - ■ ■ ■ Billing / workflow
 - ■ ■ ■ AI clinic automation
- ■ ■ ■ FMD Academic – dental education & research workspace
 - ■ ■ ■ AI seminar / PPT Studio
 - ■ ■ ■ Literature / research assistant
 - ■ ■ ■ Journal Club
 - ■ ■ ■ Case presentation
 - ■ ■ ■ Poster generation
 - ■ ■ ■ Viva / presentation rehearsal

One account should allow a user to move between products without creating unrelated identities.

3. FMD Main — Public Dental App

FMD Main is the consumer-facing dental ecosystem. It should feel premium, modern and socially native—not like a traditional healthcare directory.

Core modules

- **AI Dental Assistant:** patient describes a dental problem by text or voice; AI explains possibilities, urgency, relevant specialty and next steps without pretending to make a definitive diagnosis.
- **Dentist profiles:** Instagram-like professional profiles with credentials, specialties, clinic information, content, posts and reels.
- **Explore Content:** personalized dental content based on the user's question/problem.
- **Discovery & booking:** find relevant dental professionals/clinics and book appointments.
- **Digital records:** patient-controlled storage for X-rays, photographs, treatment documents and history.
- **Digital warranty:** treatment/warranty certificates where clinically and legally appropriate.
- **Dental community:** posts, reels, stories, polls and jobs; no live streaming in the initial scope.
- **Dental tourism:** future international patient discovery, agents/partners and treatment coordination.

Patient AI workflow

User input → language/voice detection → dental intent extraction → safety/triage layer → AI explanation → likely area/specialty → relevant content → nearby/selected dentist discovery → booking → record continuity.

4. FMD Clinic — Practice Management

FMD Clinic should operate as the clinic-side system connected to FMD Main.

- Appointment calendar and chair/doctor scheduling
- Patient registration and profiles
- Automated reminders and confirmations
- Treatment records, photos and X-ray uploads
- Billing and payment workflow
- Treatment plan communication
- Patient follow-up and recall
- AI-assisted appointment responses and clinic messaging
- Dashboard for clinic performance and operational metrics
- Future integrations: payment gateways, EMI/NBFC providers and permitted third-party services

The platform should keep treatment fees and money flows architecturally separated from any platform fee/commission model unless a reviewed legal/compliance structure explicitly permits otherwise.

5. FMD Academic — Flagship AI Academic Workspace

FMD Academic should be a dedicated experience in the FMD ecosystem, with the polish of a standalone academic product.

Positioning

FMD Academic — The AI academic workspace for dentistry.

Primary capabilities

- **AI Seminar Studio:** create 30–150+ slide dental seminars.
- **Specialty-aware generation:** the same topic should produce different academic emphasis for Prosthodontics, Periodontics, Orthodontics, Oral Surgery, etc.
- **Literature research:** search, retrieve, summarize and cite current + landmark literature.
- **Case Presentation:** structured clinical case decks.
- **Journal Club:** paper summary, critical appraisal, discussion points and viva questions.
- **Poster Studio:** academic poster generation.
- **Research Assistant:** literature synthesis, evidence tables and research planning.
- **Viva AI:** faculty-style questioning based on the exact presentation.
- **Speaker notes:** presenter-ready notes for every slide.

6. FMD Academic Studio — Required AI Workflow

Example: MDS Prosthodontics student enters “Calcium Metabolism — Seminar — 100 slides — 45–60 minutes — Vancouver references.”

```
USER INPUT
↓
TOPIC ANALYZER
↓
SPECIALTY CONTEXT ENGINE
↓
ACADEMIC OUTLINE / BLUEPRINT
↓
LITERATURE RESEARCH
↓
FMD DENTAL KNOWLEDGE RETRIEVAL
↓
SLIDE ARCHITECT
↓
CONTENT GENERATION
↓
VISUAL / DIAGRAM PLANNER
↓
CITATION VERIFICATION
↓
ACADEMIC QUALITY REVIEW
↓
FMD DESIGN / PPT ENGINE
↓
PPTX + PDF + SPEAKER NOTES + REFERENCES
↓
VIVA MODE
```

The system must not generate 100 slides in one uncontrolled LLM call. It should first create and validate a master blueprint, then generate structured slide objects in controlled batches.

7. Benchmark: Calcium Metabolism + MDS Prosthodontics

This should be one of the first benchmark presentations used to evaluate the system.

Expected academic progression

- Introduction and learning objectives
- Basic calcium physiology and distribution
- Serum calcium: ionized vs protein-bound
- Intestinal absorption and renal handling
- PTH, vitamin D, calcitonin, FGF23 and calcium-sensing receptor
- Calcium-phosphate relationship
- Bone cells and bone remodeling
- RANK/RANKL/OPG and mineralization
- Hypocalcemia, hypercalcemia and relevant metabolic disorders
- Osteoporosis and systemic factors affecting bone
- Alveolar bone and residual ridge resorption
- Implant osseointegration and bone healing
- Calcium-phosphate biomaterials and graft-related concepts
- Clinical implications for prosthodontic patients
- Recent literature, controversies and future directions
- Conclusion and clinical takeaways

The presentation must feel like an MDS Prosthodontics seminar, not a generic medical/biology PPT with a few dental slides appended.

8. Specialty Intelligence Layer

Create a structured specialty ontology/knowledge graph so that the AI understands how each specialty approaches the same topic.

Specialty	Example emphasis
Prosthodontics	Occlusion, dentures, implant prosthodontics, biomaterials, esthetics, edentulous ridge, digital workflow
Periodontics	Periodontal tissues, inflammation, regeneration, bone loss, mucogingival considerations, implant maintenance
Orthodontics	Growth, biomechanics, tooth movement, anchorage, craniofacial development
Oral Surgery	Surgical anatomy, indications, complications, healing, anesthesia, perioperative management
Conservative / Endodontics	Pulp biology, caries, adhesive dentistry, irrigation, obturation, restorative outcomes
Oral Medicine	Systemic disease, oral manifestations, diagnosis, differential diagnosis, medical management
Oral Pathology	Histopathology, molecular mechanisms, differential diagnosis, disease classification
Pediatric Dentistry	Growth, development, behavior, prevention, primary dentition, special needs
Public Health Dentistry	Epidemiology, prevention, programs, access, population-level evidence

9. Dental Knowledge Base + Research Layer

The AI should not rely solely on general model memory. FMD needs a controlled dental knowledge and retrieval layer.

Knowledge organization

Specialty → Subject → Topic → Subtopic → Source → Evidence type → Date → Metadata

Potential source categories

- Public-domain / licensed / authorized textbooks and educational resources
- PubMed-indexed literature and metadata
- Crossref / DOI metadata
- Systematic reviews and meta-analyses
- Professional guidelines and consensus statements
- Institutionally permitted resources
- FMD-created educational content

Legal requirement: do not scrape or reproduce copyrighted books/journal full text without appropriate rights. Retrieval should respect publisher/API licenses and copyright limits.

10. FMD Presentation / PPT Engine

The presentation engine should be separate from the AI reasoning layer.

Recommended architecture

LLM / Research Layer ↓ Structured Slide JSON ↓ FMD Template Engine ↓ PPTX Renderer ↓ Editable PowerPoint (.pptx) ↓ PDF export

Build approximately 25–30 high-quality master templates initially:

- Title / section divider / learning objectives
- Definition / classification / timeline
- Mechanism / biological pathway / flowchart
- Comparison / evidence table / research paper
- Clinical case / clinical algorithm / treatment workflow
- Image + explanation / anatomy / histology
- Chart / data visualization / key takeaway
- Summary / conclusion / references

The AI should select the appropriate visual template based on the content. The output should be editable and consistent, not a collection of flattened images.

11. Structured Slide Object

Every generated slide should exist as structured data before rendering.

```
{ slide_number: 77, slide_type: "mechanism", title: "Calcium Metabolism and Implant Osseointegration",  
  purpose: "Connect calcium biology with prosthodontic implant healing", key_points: [...],  
  visual_requirement: "mechanism_diagram", specialty_relevance: "...", citations: [...], speaker_notes:  
  "...", confidence: 0.92 }
```

This allows the system to regenerate one slide, change a theme, update citations, or export to another format without regenerating the entire presentation.

12. Academic Quality-Control System

Before delivery, the presentation should pass an automated review and, for flagship benchmarks, expert human review.

QC dimension	What to check
Scientific accuracy	Claims, mechanisms, terminology and clinical statements
Academic depth	Appropriate for selected BDS/MDS/faculty level
Specialty relevance	Topic is meaningfully connected to selected specialty
Completeness	Important concepts and recent developments are not missing
Duplication	Slides do not repeat the same information
Evidence	Major scientific claims have appropriate sources
Citation validity	DOI/PubMed/source metadata is valid
Visual quality	Text density, hierarchy, diagrams, tables and readability
Narrative	Basic → advanced → clinical → current evidence → conclusion

Create an internal benchmark suite of dental topics and have MDS faculty score outputs. Use those scores to improve prompts, specialty rules, retrieval and templates.

13. FMD Main — AI Dental Assistant

The public AI should be useful while maintaining appropriate clinical safety boundaries.

Example workflow

Patient: "My lower back tooth hurts when I drink cold water. Explain in Telugu."

- language detection
- intent extraction
- symptom education / possible causes
- urgency/safety check
- relevant specialty
- personalized content
- dentist discovery
- booking
- record continuity

Potential capabilities

- Text + voice interaction
- Multilingual support including Indian languages
- Document/photo/X-ray intake where appropriate
- Personalized content recommendations
- Specialist routing
- Clinic appointment assistance
- AI-generated non-diagnostic educational explanations

Medical safety, privacy, consent, disclaimers, human escalation and data governance must be designed into the system rather than added later.

14. FMD Social / Professional Layer

The public app should make professional dentistry feel modern and engaging.

- Instagram-style dentist profiles
- Posts, reels, stories and polls
- Dental content discovery
- Professional jobs
- Creator profiles
- Clinic profiles
- Verified professional credentials where supported
- Content moderation and reporting
- Future creator/brand partnerships

The social layer should remain dentistry-first rather than becoming a generic social network.

15. Payments, Booking & Monetization Architecture

Monetization should be modular and legally reviewed before launch.

Stream	Concept
Patient/public	Core consumer access can remain free or freemium
Professional membership	Low-cost annual membership for dentists/students as planned
Clinic SaaS	Monthly clinic management subscription
Academic	Potential student/professional subscription for advanced Academic features
Advertising	Relevant dental/healthcare advertising subject to policy and law
Sponsored content	Clearly disclosed sponsorships
Featured visibility	Paid visibility options with transparent rules
Partnerships	Banks/cards, dental brands, education and other permitted partners
EMI / financing	Integrate licensed lenders/payment partners where legally appropriate

Any treatment-fee, referral, commission or payment-splitting mechanism must undergo dedicated Indian legal/compliance review before implementation.

16. Security, Privacy & Compliance by Design

- Role-based access control for patient, dentist, clinic staff, student, faculty and admin.
- Encryption in transit and at rest.
- Strong authentication and session management.
- Audit logs for sensitive record access.
- Explicit consent for storage/processing of health information.
- Data minimization and retention policies.
- Secure object storage for X-rays/photos/documents.
- Tenant isolation for clinic data.
- Backups and disaster recovery.
- AI logging that avoids unnecessary retention of sensitive data.
- Clear clinical disclaimers and human escalation pathways.
- India-specific privacy, healthcare, advertising and professional-regulation review before production.

17. FMD Super Admin / Control Center

The super-admin system should control the ecosystem centrally.

Area	Controls
Users	Patients, dentists, students, faculty, clinics, institutions
Verification	Professional credentials / clinic verification workflows
Content	Moderation, reports, takedowns, featured content
AI	Prompt/version management, model routing, usage limits, evaluation
Academic	Knowledge sources, specialty ontology, templates, benchmark results
PPT	Template library, generation jobs, failures, quality scores
Clinics	Subscriptions, clinic status, usage and support
Payments	Transactions, subscriptions, refunds and reconciliation
Analytics	Acquisition, engagement, retention, conversion, revenue
System	API health, queues, storage, security events, errors

18. Recommended Technical Architecture

The implementation should be modular so FMD can swap AI models or external services without rewriting the product.

MOBILE / WEB CLIENTS ↓ API GATEWAY / AUTH ↓ FMD CORE SERVICES ■■■ User & Identity ■■■ Profiles / Social ■■■ Booking ■■■ Clinic ■■■ Payments ■■■ Content ■■■ Records ■■■ Academic ■■■ Notifications ↓ AI ORCHESTRATION LAYER ■■■ Model router ■■■ Prompt/version manager ■■■ RAG / retrieval ■■■ Research connectors ■■■ Safety layer ■■■ Evaluation / QC ■■■ Usage / cost controls ↓ DATA LAYER ■■■ PostgreSQL ■■■ Vector search ■■■ Object storage ■■■ Cache ■■■ Analytics warehouse ↓ EXTERNAL SERVICES ■■■ LLM providers ■■■ PubMed / scholarly APIs ■■■ Payment gateway ■■■ Messaging / email / SMS ■■■ Optional PPT renderer / PDF services

Use APIs and managed infrastructure wherever practical. FMD's value is the orchestration, domain intelligence, UX, data model, evaluation and ecosystem—not recreating foundational AI infrastructure.

19. Non-Functional Requirements

- **Scalability:** design for independent scaling of AI jobs, media, booking and social services.
- **Reliability:** queued background jobs for long AI/PPT generations; retry and recovery.
- **Observability:** logs, traces, metrics and AI generation telemetry.
- **Cost control:** model routing, token budgets, caching, retrieval limits and generation quotas.
- **Versioning:** version prompts, specialty rules, templates and AI evaluation sets.
- **Editable output:** PPTX must remain editable.
- **Localization:** architecture should support Indian languages.
- **Accessibility:** readable typography, captions/alt text where appropriate and keyboard/accessibility support for web.
- **API-first:** core services should be reusable by mobile apps, web and future FMD products.

20. Build Strategy — Killer Product From Day One

“Killer from day one” should mean a strong architecture and polished experience, not building every future feature simultaneously.

Phase	Priority
Foundation	FMD identity, core backend, AI orchestration, security, analytics and shared design system
FMD Main	AI assistant, profiles, content, booking, records and core social experience
FMD Clinic	Appointments, patient management, reminders, billing/workflow and clinic dashboard
FMD Academic	Academic Studio architecture, specialty intelligence, research/RAG, PPT engine, Viva
Quality	Benchmark suite, faculty evaluation, AI QC and model evaluation
Scale	Internationalization, dental tourism, institutions, partnerships and additional AI agents

Recommended first Academic benchmark: Calcium Metabolism — MDS Prosthodontics — ~100 slides. If this is genuinely faculty-level, the architecture is working.

21. Developer Acceptance Criteria

The development team should be able to demonstrate the following before calling the core Academic system production-ready:

- A student can enter topic + specialty + level + duration + slide count and receive a structured academic blueprint.
- The blueprint can be reviewed/edited before full generation.
- The system can retrieve relevant literature and retain source metadata.
- Every slide has structured data and can be regenerated independently.
- At least 25–30 professional slide templates exist.
- PPTX export is editable and visually consistent.
- Citations are attached to claims and validated where technically possible.
- Academic QC identifies duplication, unsupported claims and missing sections.
- Specialty context materially changes the presentation.
- Speaker notes and reference list are generated.
- Viva mode is generated from the exact presentation.
- Generation jobs can survive timeouts/retries.
- Usage, cost and failure metrics are visible to super admin.

22. Future FMD Intelligence Layer

The long-term opportunity is to build a reusable FMD intelligence layer powering multiple dental workflows.

Agent	Purpose
Patient AI	Dental education, triage and discovery
Clinic AI	Scheduling, communication and operational automation
Academic AI	Seminars, presentations, literature and viva
Research AI	Literature synthesis, evidence tables and research workflows
Content AI	Dental posts, reels, captions and professional content
Voice AI	Patient/clinic/academic voice interaction
Future Campus AI	Institutional academic/admin workflows, if/when activated

The products should share infrastructure where appropriate but retain separate UX and permission boundaries.

23. One-Sentence Product Definition

FMD is a dentistry-first technology ecosystem that combines AI, dental discovery, professional identity, clinic operations, patient continuity, education and research into one connected platform.

For the development team:

Build the architecture so FMD can evolve from a public dental app into a family of specialized products—without having to rebuild the core platform each time a new FMD experience is introduced.

Immediate technical discussion checklist

- Confirm product boundaries: Main / Clinic / Academic.
- Confirm shared identity and backend architecture.
- Select primary AI provider and model-routing strategy.
- Design FMD dental RAG / knowledge architecture.
- Design Academic Studio slide JSON + template engine.
- Estimate PPT generation infrastructure and per-generation cost.
- Design security/privacy model before handling patient records.
- Define API contracts and database schema.
- Build the Calcium Metabolism benchmark.
- Create development milestones, budget and timeline.

This document is a product/technical direction document. Final architecture, legal/compliance controls, vendor selection and production specifications should be validated by the development, security and legal teams before implementation.